R16

Code No: 135CU

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
NON-CONVENTIONAL POWER GENERATION
(Common to CE, ME, ECE, CSE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What are solar collectors? Compare different types of solar collectors on the basis of construction and application.
b) Explain the principle of conversion of solar energy into heat. [8+7]

2. Discuss with sketch and equations how solar radiation measurement is calculated on a horizontal surface. [15]

3.a) Discuss about the main elements of a PV system and its Equivalent Circuit.
b) Draw the V-I characteristics of PV system and explain. [8+7]

4.a) Define Maximum Power Point Tracking(MPPT) and explain its significance.
b) Enumerate the design parameters to be considered to implement MPPT technique. [7+8]

5. What is the total power of a wind stream? What do you mean by coefficient of performance and tip speed ratio of a windmill? On what factors does the performance of windmills depend? [15]

6. How do vertical shaft windmills compare with windmills having horizontal shaft? Explain the use of windmills. [15]

7.a) List out the Biomass resources. What are biofuels?
b) What do you understand by thermochemical and biochemical conversions? [7+8]

8.a) State the merits and demerits of OTEC plants. Explain the major problems that need to be solved.
b) How does a fuel cell operate? How is it different from a battery? Explain the hydrogen-oxygen and carbon-oxygen fuel cells. [7+8]

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